

Jayanth Utukuri

Nashua, NH | jayanthutukurius@gmail.com | +1 (603) 557-5250 | linkedin.com/in/utukuri-jayanth-20871a17b |
github.com/jayanth2909 | jayanth2909.github.io/Portfolio

Professional Summary

Full-Stack Software Engineer with **2+ years of experience** building scalable, secure, and cloud-native applications. Primary expertise in **Java 17/21, Spring Boot 3, React, and WPF (.NET 5)**, with a strong focus on UI engineering and microservice integration. Solid foundation in **data structures, algorithms, OOD, and design patterns**. Proven success delivering **microservices, REST APIs, event-driven systems, and desktop emulators** with measurable impact (**99.9% uptime, 40% latency reduction, 20% bug reduction**). Skilled in **end-to-end feature ownership** — from requirements to testing and cloud deployment — emphasizing **clean architecture and maintainable codebases**.

Technical Skills

Languages: Java 17/21, TypeScript, JavaScript (ES6+), SQL (PostgreSQL, MySQL), C#, HTML5, CSS3

Frameworks: Spring Boot 3, React, Redux, React Query, Bootstrap, WPF (.NET 5)

Microservices & APIs: REST, OpenAPI/Swagger, Kafka (event-driven), JWT, OAuth2

Cloud: AWS (EC2, S3, RDS, Lambda, CloudWatch)

DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, Terraform

Testing: JUnit 5, Mockito, Jest, Cypress, Selenium, xUnit

Practices: Agile/Scrum, CI/CD, OOD, Code Reviews, Observability, Logging

Databases: PostgreSQL, MySQL, Redis

Version Control: Git, GitHub

Work Experience

Software Engineer Intern | Tern Computer Inc., NY

May 2025 – Present

- Built a modular **C# WPF desktop emulator** simulating CPU components (ALU, Program Counter, Memory, Instruction Register) for the Balanced Ternary Machine Code (BTMC) 2024 system.
- Designed the full simulator architecture using the **MVVM pattern**, ensuring testability, scalability, and clean separation between View, ViewModel, and Model layers.
- Created custom **WPF UserControls and CustomControls** (RegisterField, GeneralPurposeRegisterControl, MemoryControl, SimulatorConsole) with advanced styling via **ResourceDictionary**.
- Used **Dependency Properties, Bindings, and Commands** for real-time synchronization between CPU engine logic and UI components.
- Implemented multithreaded instruction execution with **Dispatcher.Invoke** and **Task.Run**, ensuring the UI remains responsive during simulation.
- Integrated **DataTemplates, GridView, and ObservableCollection** for dynamic rendering of 27+ registers and 64 memory cells, achieving pixel-perfect alignment per Figma specifications.
- Applied **UI virtualization and async data binding** to improve performance, reducing UI rendering lag by **35%**.
- Developed robust validation mechanisms using **WPF ValidationRules** and utility-based regex patterns for input correctness.
- Engineered a modular **Arithmetic Logic Unit (ALU)** supporting core ternary operations (ADD, SUB, AND, OR, SLT) with trace logging and error handling.
- Integrated the emulator with backend **Spring Boot REST APIs** for analytics and state persistence, secured with JWT-based authentication.
- Added structured **debug console logging** in WPF for cycle-by-cycle tracing of instruction flow (Fetch–Decode–Execute–Writeback).
- Collaborated in Agile sprints, writing detailed PRs, QA validation notes, and achieving **90%+ xUnit test coverage**.

Graduate Research Assistant | Rivier University, Nashua, NH

May 2025 – Jun 2025

- Supported a **computer vision research project**, focusing on image data collection, preprocessing, and model training.
- Improved model accuracy by **10%** through **feature extraction, normalization, and hyperparameter tuning**.
- Utilized **Python, OpenCV, NumPy, and scikit-learn** for data manipulation and model development.
- Conducted literature reviews and maintained detailed experiment documentation to ensure reproducibility.

Graduate Teaching Assistant | Rivier University, Nashua, NH

Jan 2025 – Apr 2025

- Mentored **30+ students** in **Java programming**, Object-Oriented Programming (OOP), debugging techniques, and software design principles.
- Designed automated grading tools in **Java**, reducing manual grading effort by **40%**.

Software Developer | Sterlite Digital Pvt Ltd, Bengaluru, India

Jul 2022 – Aug 2023

- Developed **Spring Boot 3 (Java 17)** microservices with **JPA/Hibernate** for patient data and billing, reducing deployment errors by **25%**.
- Built scalable **REST APIs** and applied **design patterns** to improve backend throughput by **28%**.
- Developed interactive **React dashboards** using **Hooks, Context API, and React Router**, improving usability and reducing UI complexity.
- Implemented **Redux Toolkit + RTK Query** for state and data management, improving performance by **35%**.
- Integrated **Spring Boot + React** stack for unified data flow and seamless front-back synchronization.
- Enhanced backend performance using **PostgreSQL indexing + Redis caching**, reducing query latency by **40%**.
- Implemented secure workflows with **JWT + OAuth2**, ensuring HIPAA compliance.
- Delivered features with **end-to-end ownership** from design through AWS deployment, improving delivery speed by **35%**.

Software Engineer Intern | Cognizant, India (Sports Platform)

Feb 2022 – Jun 2022

- Built **Spring Boot (Java 17)** microservices for live score updates and event tracking, reducing downtime by **20%**.
- Developed real-time **React dashboards** for live match updates using **WebSockets + React Hooks**.
- Implemented global state handling using **Redux + Thunk**, ensuring synchronized updates across widgets.
- Improved rendering speed by **25%** through component memoization and optimized React lifecycle usage.
- Deployed on **AWS EC2** with **Jenkins + Docker** pipelines, achieving **99.9% uptime**.
- Conducted automated testing using **JUnit, Jest, Cypress**, improving release stability by **30%**.

UI/UX Designer Intern | Configore, Hyderabad, India

Jun 2021 – Aug 2021

- Designed interactive **React UI prototypes** to improve user engagement and visual consistency.
- Collaborated with developers to translate wireframes into responsive, production-ready components.

Mobile Application Intern | The Sparks Foundation, Hyderabad, India

May 2021 – Jun 2021

- Developed cross-platform mobile apps using **React Native + Firebase**, achieving a **98% crash-free rate**.
- Implemented **modular components and hooks**, improving scalability and reusability.

Projects

AI-Driven Driver Drowsiness Detection — Built using **Keras/TensorFlow + OpenCV**, achieved **92% accuracy**, deployed via **Flask REST API + Docker**.

Real-Time Polling Platform — Developed with **React + Flask + Redis**, scaled to **1K+ concurrent users** with **99.8% uptime**. Added **React Query** and CI automation with **GitHub Actions**.

Cross-Platform Music App (Prototype) — Created in **React Native + Firebase**, achieved **<300 ms** playback latency, enabled offline mode.

Automatic Job Application Tracker (In Design) — Architecting a full-stack app with **React + Spring Boot** and a **Python AI model**. Planned **Redis caching + PostgreSQL** for storage, **Gmail API** for parsing, and dashboard refresh every 10 min.

Education

M.S., Computer Science (GPA: 4.0)

Rivier University, Nashua, NH

Sep 2023 – May 2025

B.Tech, Computer Science & Data Science

Koneru Lakshmaiah University, Hyderabad, India

Jun 2018 – Apr 2022